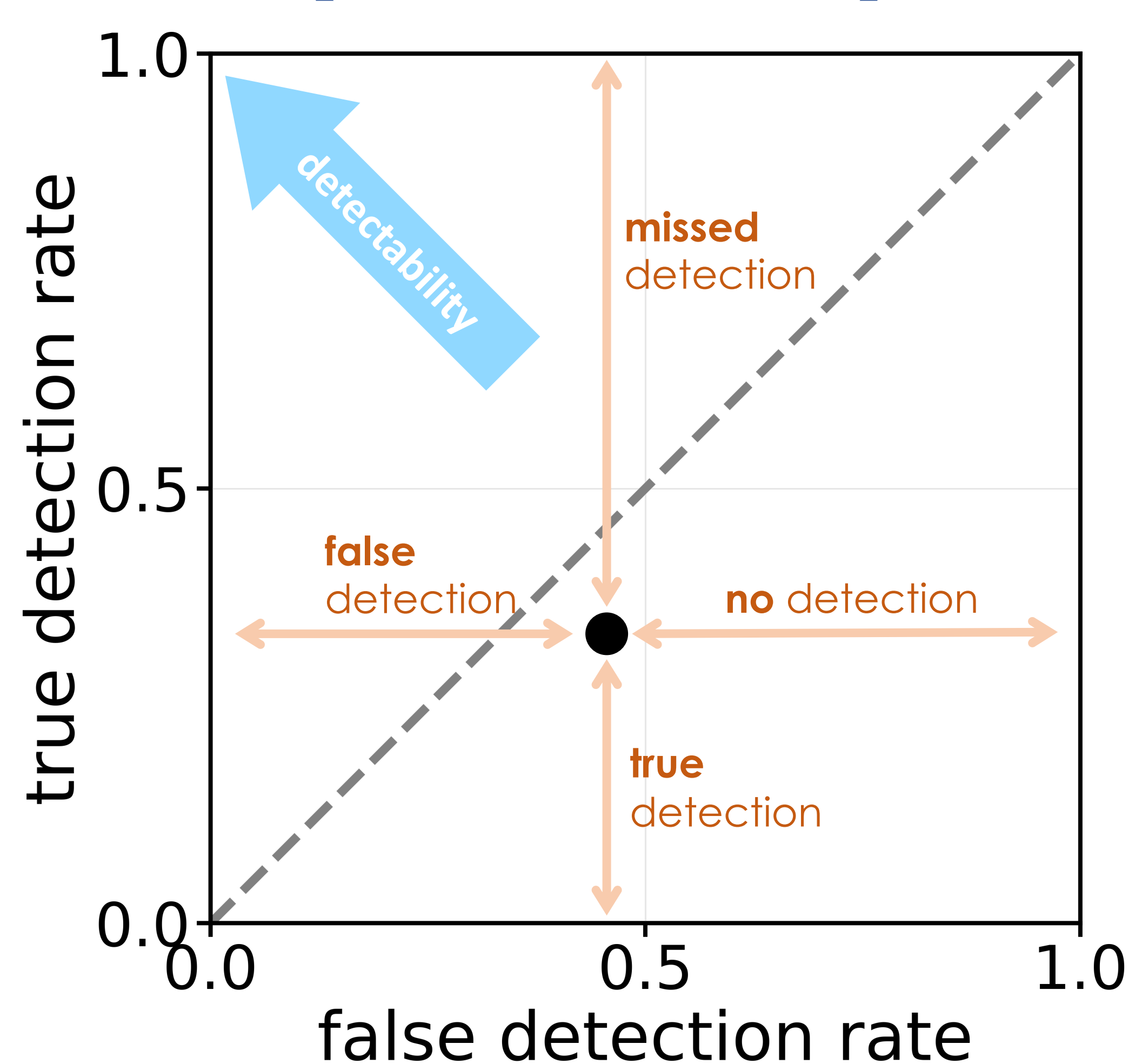
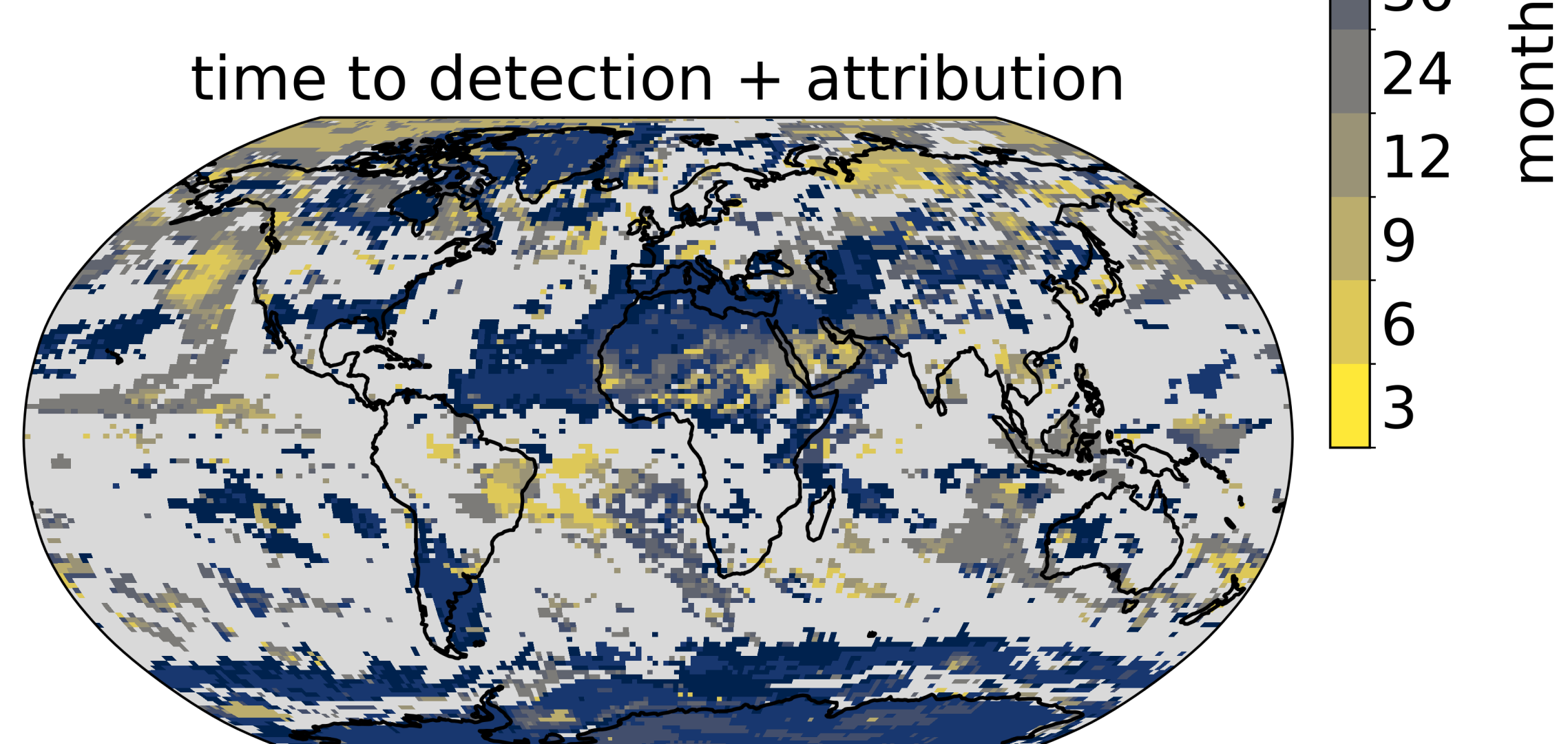
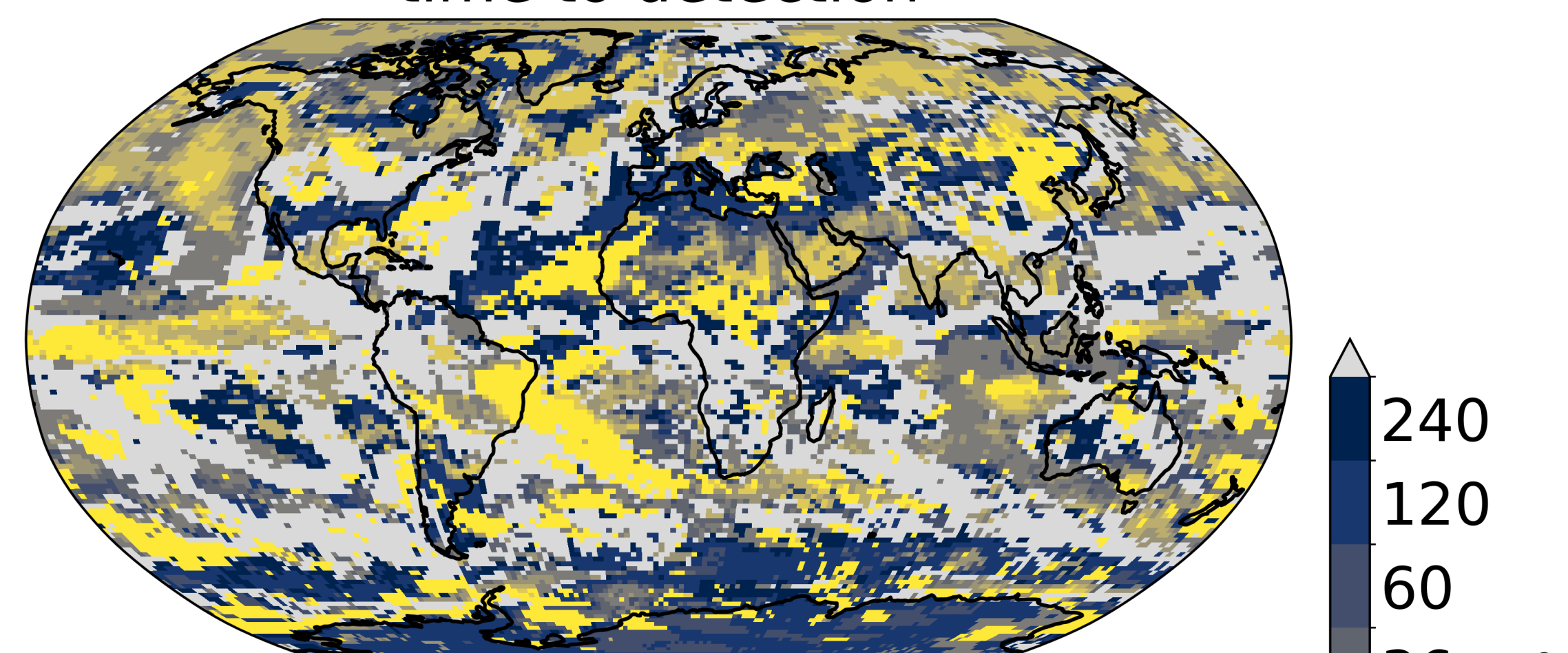
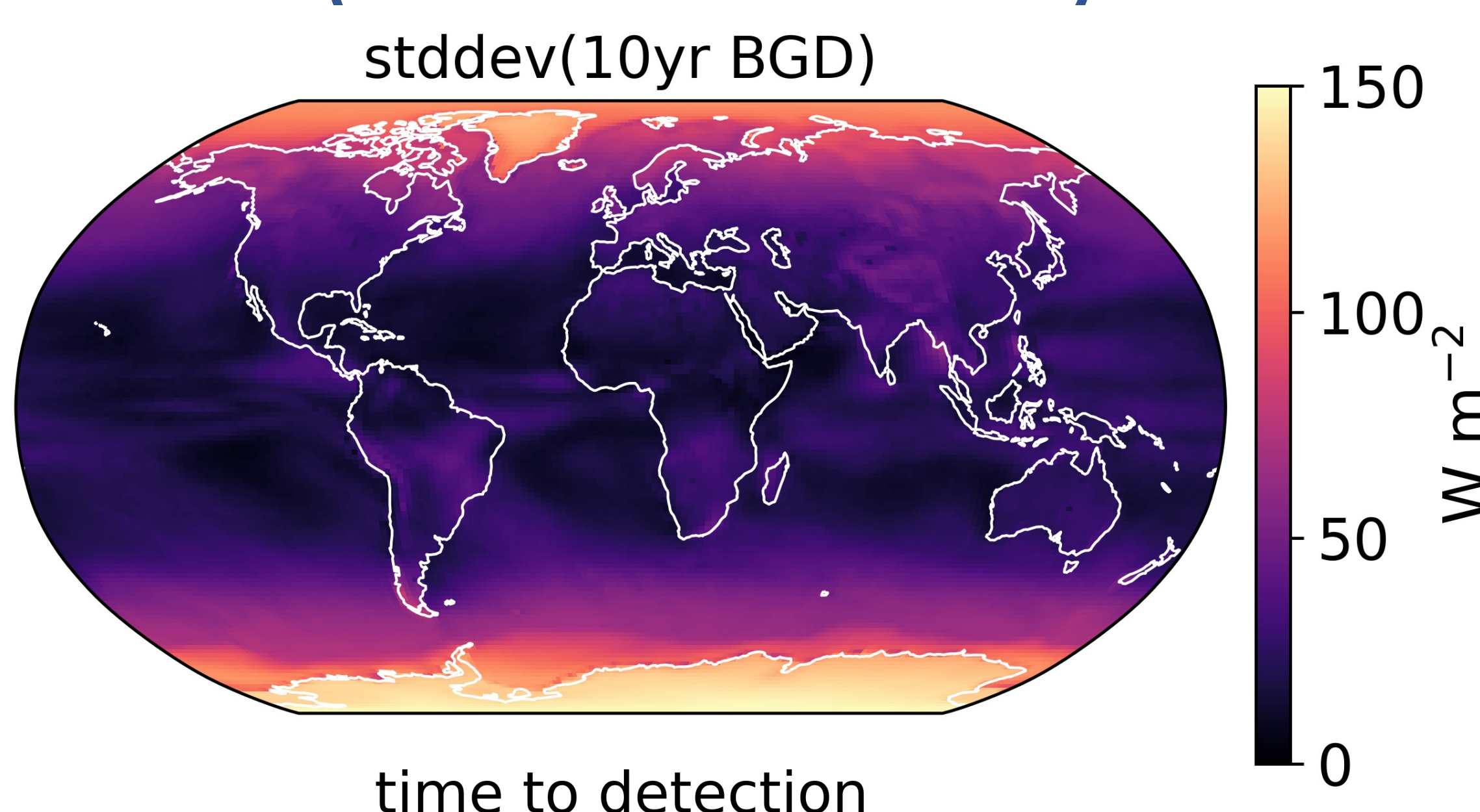




'Detectability' with Detection & Attribution (TDR vs FDR)



Spatial Patterns (in UKESM1-1)

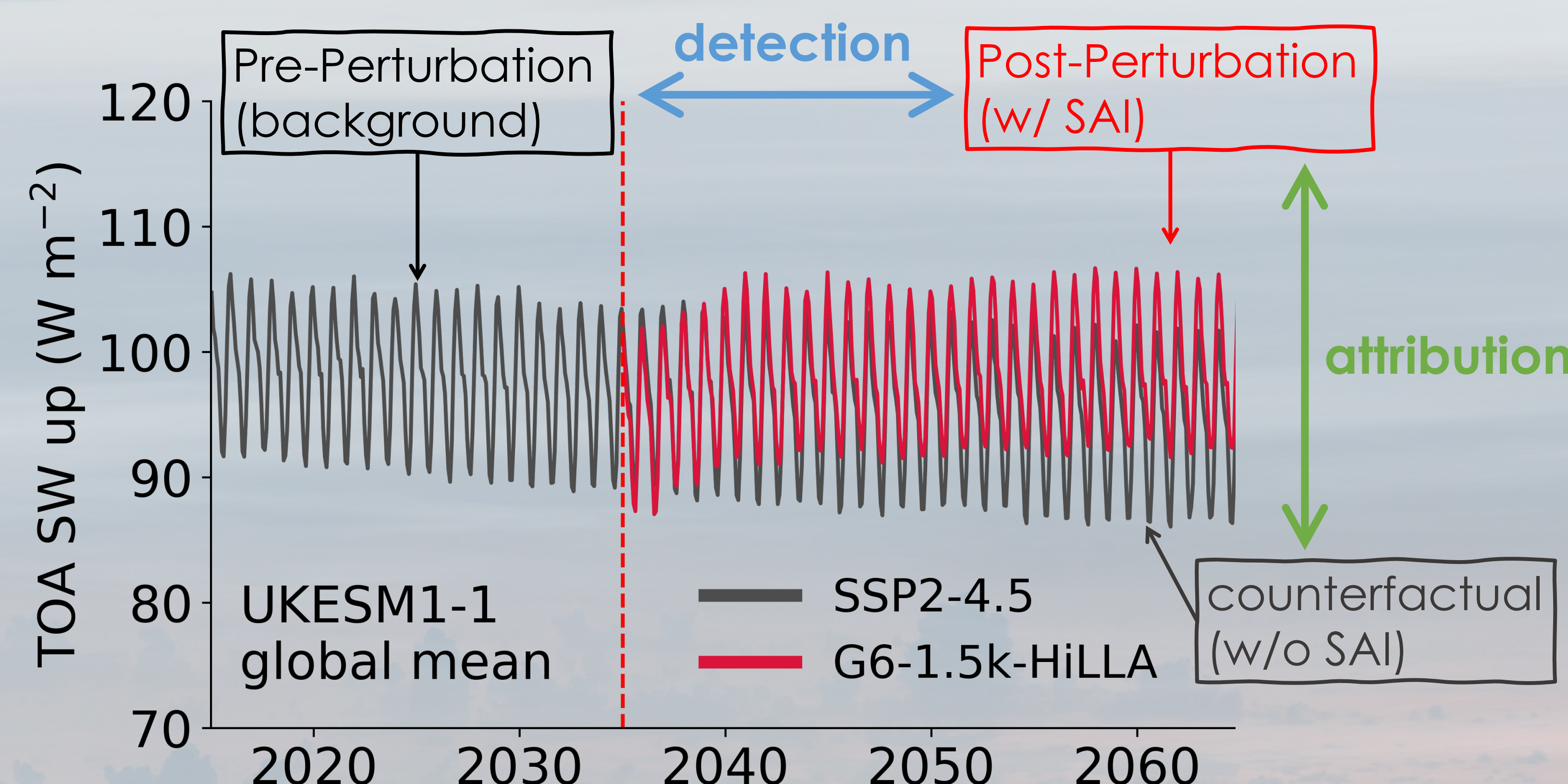


A framework for SRM detectability (detection & attribution): The case of SAI

Jianhao Zhang^{1,2} and Graham Feingold²

Takeaways:

1. Less than half of the globe is detectable, requiring on average **10 years**.
2. The **high latitude SAI** scenario increases detectable area and shortens average time to detectability.
3. **Model spread** is non-negligible, especially regionally.



Models:

CESM2-WACCM, E3SMv3, UKESM1-1

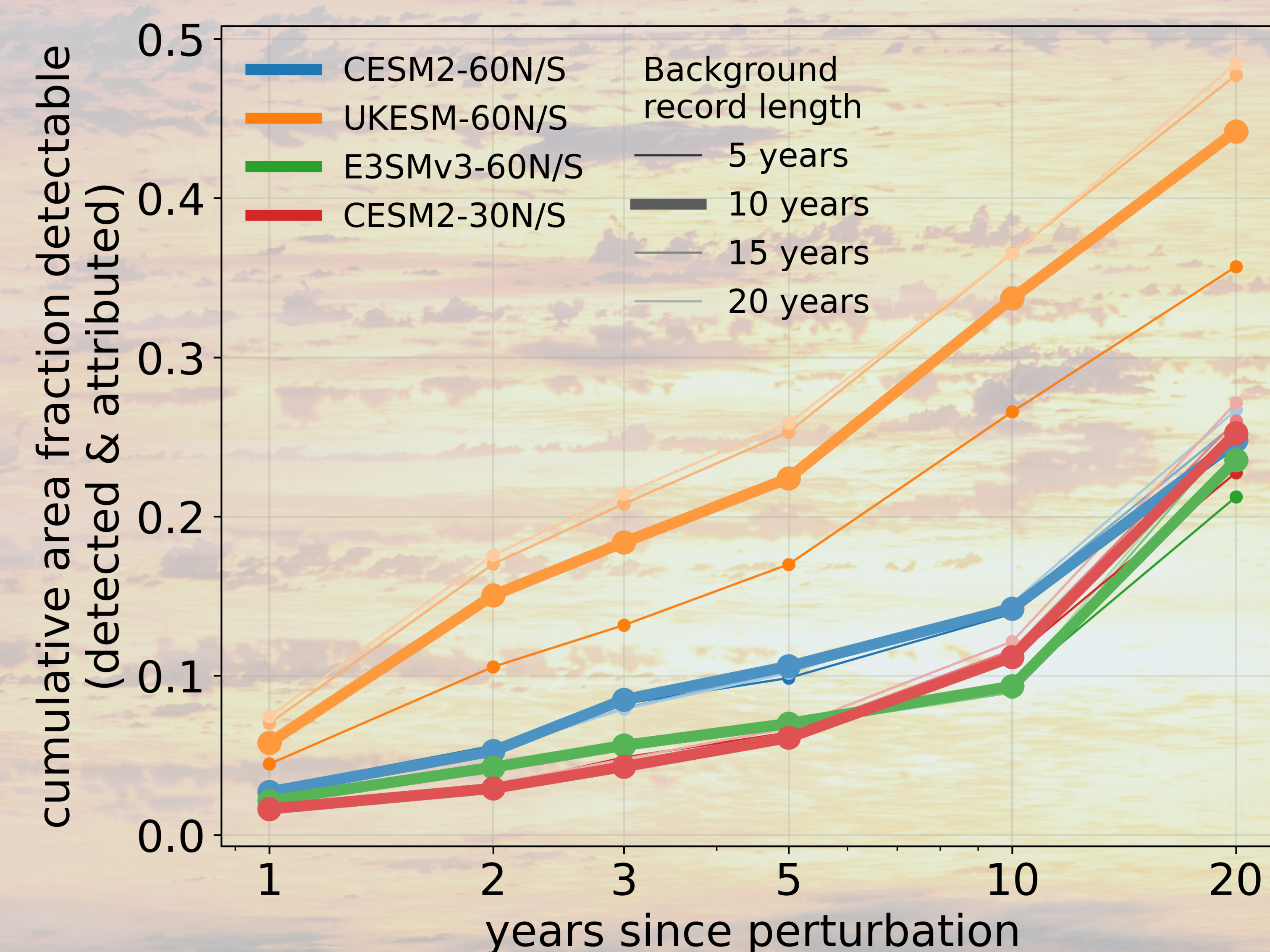
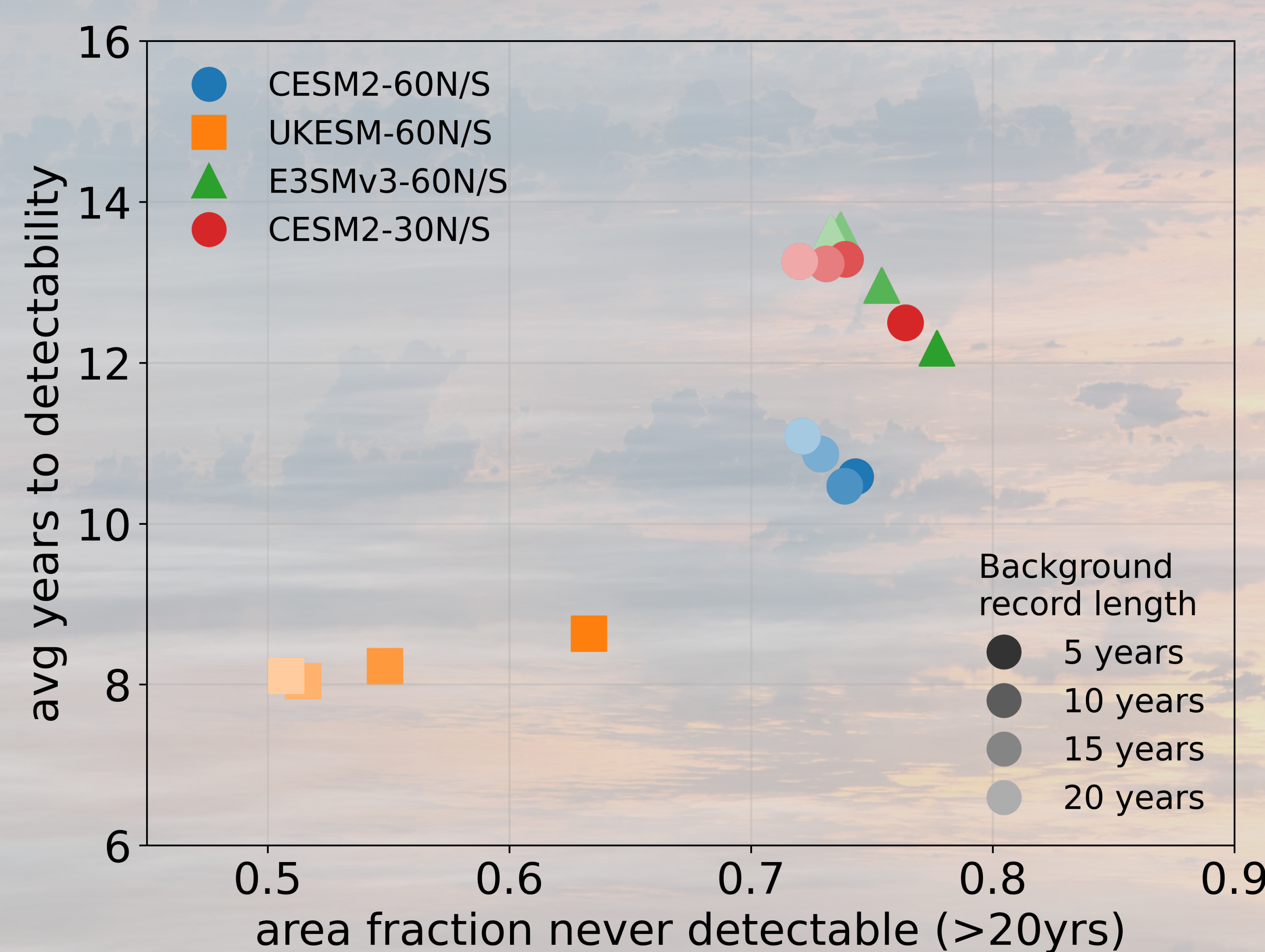
Scenarios:

(both branched from SSP2-4.5 at 2035)

- G6-1.5K-SAI (30°N/S, 21km);
- G6-1.5K-HiLLA (60°N/S, 15km)

Target variable:

SW reflection at TOA (monthly, 1°)



Zhang, J. and Feingold, G. A framework for assessing SRM detectability: Integrating detection & attribution. *In prep.*

Regional Detectability

1-year after SAI with a 10-year background

